# Lucas Zheng

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## Education

**Brown University** Providence, RI
*Sc.M. & B.S. in Computer Science, B.A. in Applied Mathematics; GPA: 4.00/4.00* *Aug. 2023 – May 2028*

- Coursework: Probability & Statistics, Machine Learning, Deep Learning, Sequential Decision Making, Algorithms, Applied PDEs

## Experience

**Applied Machine Learning Intern** Jun. 2026 – Present
*Shopify* *New York, NY*

- Built an end-to-end, locale-aware evaluation system gating AI-generated sales-outreach emails, spanning labeled eval-set generation, LLM-judge scoring, and a feedback loop turning reviewer annotations into prompt revisions
- Extended the gate to 6 non-US locales covering 415K outreach emails per year, improving judge F1 against human-labeled ground truth from 0.57 to 0.74 through model and context-rotation upgrades

**Research Assistant** Sep. 2025 – Present
*Intelligent Robot Lab, Brown University* *Providence, RI*

- Developed an option-level (hierarchical RL) RMax agent with prioritized-sweeping planning for Montezuma's Revenge under George Konidaris, reaching 6,800 score and 22 of 24 rooms with 28 temporally extended skills
- Built an action-conditioned Coin Flip Network for intrinsic-reward exploration atop a custom PyTorch DQN with action masking, comparing state-only and action-conditioned intrinsic rewards

**Software Engineering Intern** Jul. 2025 – Aug. 2025
*Supio* *Seattle, WA*

- Reduced deposition transcript turnaround time from 10 days to minutes by engineering an end-to-end pipeline with Whisper, pyannote speaker diarization, and an OpenAI API correction workflow
- Integrated transcripts into hybrid RAG search system combining pgvector semantic similarity with BM25 ranking algorithms, enabling lawyers to query hours of audio content with sub-second response times

**Founder** Jul. 2024 – Dec. 2025
*12M Analytics* *Providence, RI*

- Built and shipped a college football analytics platform (React/TypeScript, Flask, PostgreSQL) with a LangGraph analytics agent over coach-uploaded game data; drove adoption to 10+ collegiate programs through direct outreach

## Projects

**2026 U.S. Senate Election Forecast** | *Python, pandas, scikit-learn, TypeScript/Next.js* Jul. 2026 – Present

- Forecast all 35 races by blending a ridge fundamentals prior with a weighted poll average as a nested-CV-tuned pseudo-poll, then a 40,000-path Monte Carlo with correlated fat-tailed national, regional, and state errors
- Backtested on held-out 2018–2024 election cycles: the blend's Brier score was 0.071, beating the fundamentals prior (0.082, −14%) and polls alone (0.085, −17%)

**Physics-Informed Neural Network for Seismic Localization** | *PyTorch, Gymnasium, Stable-Baselines3* 2025

- Formulated PINN collocation-point selection as an MDP and trained a PPO agent over a 403-dim residual state and 400 discrete actions, achieving the fastest physics-loss convergence among 4 sampling strategies
- Built a PINN solving the eikonal PDE to localize seismic sources from 20 receivers over a heterogeneous velocity model, with Fourier-feature encoding to overcome MLP spectral bias

## Technical Skills

**Languages**: Python, TypeScript, SQL
**Tools & Frameworks**: PyTorch, pandas, Gymnasium, scikit-learn, PostgreSQL, MCP, Git, REST APIs

## Miscellaneous

**Hackathons**: 2nd Place winner at 2026 Precision Neuroscience BCI Hackathon
**Languages**: Mandarin (fluent), Arabic (working proficiency – U.S. State Department NSLI-Y Scholar)
**Interests**: Squash, Golf, Poker, Chess